

Basic Commands

1. `**`pwd`**`: Print Working Directory. Shows the current directory.
2. `**`ls`**`: List files and directories.
3. `**`cd`**`: Change Directory.
4. `**`cp`**`: Copy files or directories.
5. `**`mv`**`: Move or rename files or directories.
6. `**`rm`**`: Remove files or directories.
7. `**`mkdir`**`: Make Directory.
8. `**`rmdir`**`: Remove Directory (empty).
9. `**`touch`**`: Create an empty file or update a file's timestamp.
10. `**`cat`**`: Concatenate and display file content.
11. `**`echo`**`: Display text or variable values.

File Commands

1. ****`file`****: Determines file type.
 - Example: ``file file1.txt``
2. ****`chmod`****: Change file permissions.
 - Example: ``chmod 755 file1.txt``
3. ****`chown`****: Change file owner and group.
 - Example: ``chown user1 file1.txt``
4. ****`ln`****: Create hard or symbolic links.
 - Example: ``ln -s file1.txt link1.txt`` (symbolic link)
5. ****`diff`****: Compare files line by line.
 - Example: ``diff file1.txt file2.txt``
6. ****`tar`****: Archive files.
 - Example: ``tar -cvf archive.tar file1.txt``, ``tar -xvf archive.tar``
7. ****`gzip`****: Compress files.
 - Example: ``gzip file1.txt``
8. ****`gunzip`****: Decompress files.
 - Example: ``gunzip file1.txt.gz``
9. ****`zip`****: Create a zip archive.
 - Example: ``zip archive.zip file1.txt``
10. ****`unzip`****: Extract zip archive.
 - Example: ``unzip archive.zip``

System Commands

1. **top**: Display real-time system processes.
 - Example: `top`
2. **ps**: List running processes.
 - Example: `ps aux`
3. **kill**: Terminate processes.
 - Example: `kill 1234`
4. **pkill**: Kill processes by name.
 - Example: `pkill firefox`
5. **systemctl**: Manage system services.
 - Example: `systemctl status apache2`, `systemctl start apache2`
6. **service**: Manage system services (older command).
 - Example: `service apache2 status`
7. **reboot**: Restart the system.
 - Example: `reboot`
8. **shutdown**: Shut down or restart the system.
 - Example: `shutdown now`, `shutdown -r now` (restart)
9. **df**: Display disk space usage.
 - Example: `df -h`
10. **du**: Disk usage of files and directories.
 - Example: `du -sh /home/user/`

Network Commands

1. **`**`ifconfig`**`**: Display or configure network interfaces (deprecated, replaced by ``ip``).
 - Example: ``ifconfig``
2. **`**`ip`**`**: Network interface configuration.
 - Example: ``ip addr show``, ``ip link set eth0 up``
3. **`**`ping`**`**: Check connectivity to a host.
 - Example: ``ping google.com``
4. **`**`traceroute`**`**: Trace the route packets take to a network host.
 - Example: ``traceroute google.com``
5. **`**`netstat`**`**: Network statistics and connections.
 - Example: ``netstat -tuln``
6. **`**`ss`**`**: Utility to investigate sockets.
 - Example: ``ss -tuln``
7. **`**`curl`**`**: Transfer data from or to a server.
 - Example: ``curl http://example.com``
8. **`**`wget`**`**: Download files from the web.
 - Example: ``wget http://example.com/file.zip``
9. **`**`nmap`**`**: Network exploration and security auditing.
 - Example: ``nmap 192.168.1.1``
10. **`**`iptables`**`**: Configure network packet filtering rules.
 - Example: ``iptables -L`` (list rules)

Other Useful Commands

1. **`**`history`**`**: Show command history.

- Example: ``history``
- 2. **`**`alias`**`**: Create shortcuts for commands.
 - Example: ``alias ll='ls -l``
- 3. **`**`unalias`**`**: Remove aliases.
 - Example: ``unalias ll``
- 4. **`**`basename`**`**: Strip directory and suffix from filenames.
 - Example: ``basename /home/user/file1.txt``
- 5. **`**`dirname`**`**: Strip the last component from the file name.
 - Example: ``dirname /home/user/file1.txt``
- 6. **`**`date`**`**: Display or set the system date and time.
 - Example: ``date``, ``date '+%Y-%m-%d %H:%M:%S'``
- 7. **`**`uptime`**`**: Show how long the system has been running.
 - Example: ``uptime``
- 8. **`**`whoami`**`**: Show the current user.
 - Example: ``whoami``
- 9. **`**`sudo`**`**: Execute a command as another user (usually root).
 - Example: ``sudo ls /root``
- 10. **`**`man`**`**: Manual pages for commands.
 - Example: ``man man``

Additional File Management Commands#####

- 2. **`**`tree`**`**: Display directory contents in a tree-like format.
 - Example: ``tree /home/user/``
- 3. **`**`wc`**`**: Count words, lines, and characters in files.
 - Example: ``wc -l file1.txt`` (count lines in file1.txt)

Additional Network Commands

2. ****`nmap`****: Network exploration and security auditing.
 - Example: ``nmap -sP 192.168.1.0/24`` (ping scan a subnet)
4. ****`iwconfig`****: Configure wireless network interfaces.
 - Example: ``iwconfig wlan0``
5. ****`host`****: DNS lookup utility.
 - Example: ``host google.com``
6. ****`dig`****: DNS lookup and query utility.
 - Example: ``dig google.com``

Additional Miscellaneous Commands

1. ****`grep`****: Search for text patterns in files.
 - Example: ``grep "pattern" file1.txt``
6. ****`bash`****: Start a new Bash shell session.
 - Example: ``bash``
7. ****`sh`****: Start a new Bourne shell session.
 - Example: ``sh``
9. ****`sort`****: Sort lines of text files.
 - Example: ``sort file1.txt``